Useful information

- Lecturer's details.
    - Name: Steven Guan / Kok Hoe Wong / Dawei Liu
      Office: SD425 / SD431 / SD455
    - Email: steven.guan@xjtlu.edu.cn / kh.wong@xjtlu.edu.cn / dawei.liu@xjtlu.edu.cn
    - URL: Learning Mall
- TAs:
  - TBD

Overview

- Course info
- Types of computers
- Computer systems: overview
- Hardware
- Software
- Backward (Downward) compatibility
  for new hardware
- VHDL
- Hierarchy of systems
- Interacting development of hardware & software
- Hardware evolution
- Trends of computing

What is our course about?

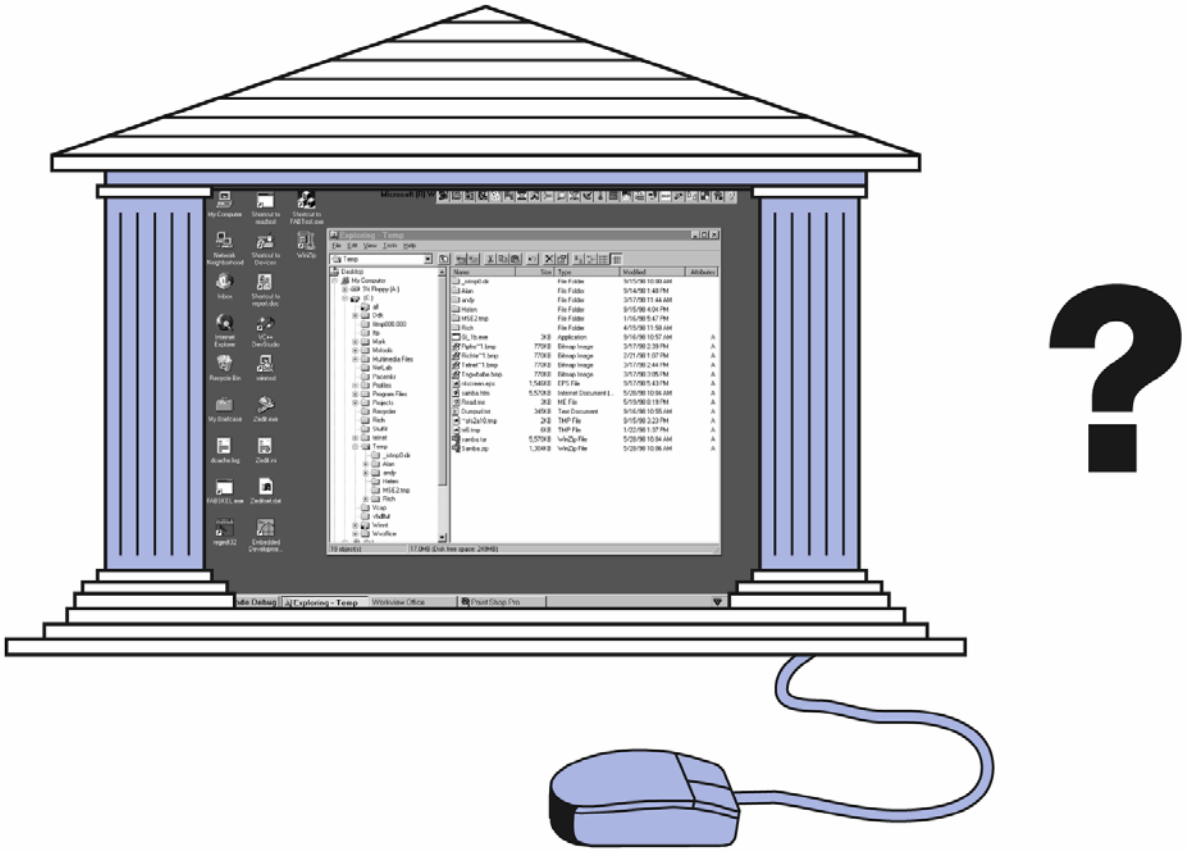

Fig. 1.1 Computer architecture.

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Aims

- To introduce students to … …
  - The components of computer systems: processors, controllers, primary memory, secondary memory, peripherals, etc.
  - The inter-working of them.

Aims (cont.)

- To introduce students to the various architecture levels of computer systems.
  - The assembly language level.
  - The instruction set architecture level.
  - The micro architecture level.
  - The digital logic level.

Learning outcome

- To understand how a computer works at the machine code level.
- To understand a little about the computer hardware.
- To understand basic principle of data storage and coding of information.
- To understand the roles and organization of the major kinds of system software.

Content of the course

- Overview of computer systems: hardware and software.
- Machine architecture and instruction execution:
  - CPU and its instruction set.
  - stored program concept, program execution, overview of instruction level operations.

Content of the course

- Machine-level correspondence to higher-level constructs:
  - Simple arithmetic expressions, loops, conditional expressions, subroutines, stack and recursions.
- Building computers from logic:
  - Elementary logic gates, simple circuits, CPU, ALU, memory.

Content of the course

- Data storage:
  - main memory, mass storage, coding of information.
- Interrupt processing – service on demand.
- Introduction to operating systems.
- Basic network concepts.

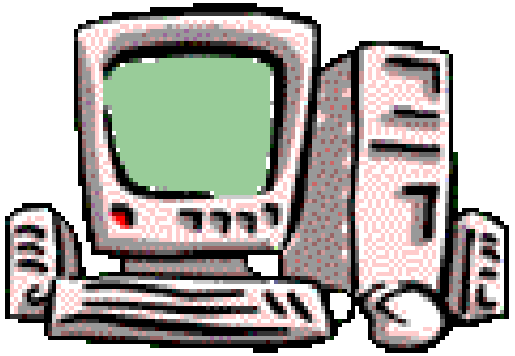
Computer Systems vs Java

- Java deals (mainly) with how programs are written at high level.

- Computer Systems deals with
  - What is going on inside the computer when the programs are run.
  - How programs are written at low level (assembly programming)

Types of Computers

- Mainframe computers (1960s)
- Supercomputers (1970s)
- Workstations (1980s)
- Microcomputers (1980s)
- Personal computers (1980s)



Types of Computers (continued)

- Microcontrollers (1980s)
  - embedded or dedicated computers: from calculators to automobiles
- Servers (1980s)
  - network
  - Web
  - Cloud
- Chip computer (?)

Increased performance ↑

↓ downsizing

Computer Generations

- First (1944 to1958): vacuum tube
  - MARK I, ENIAC and UNIVAC

- Second (1959 to1963): transistor
  - IBM1401, IBM 1410 with 1402 card read/punch

- Third generation (1964 to 1970): IC
  - DEC PDP-1
  - IBM 360

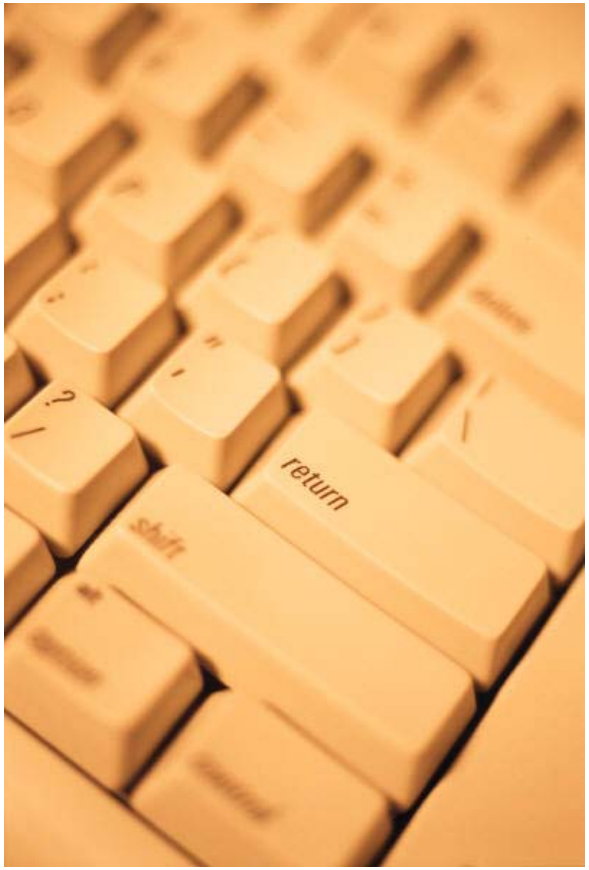
Computer Generations (continued)

- Fourth generation (1971 to now: VLSI)
  - dominated by the LSI (large-scale integrated circuits) and the VLSI (very-large-scale integrated circuits)
  - word processing, spreadsheets, database, and graphic programs became readily available
  - Cray-1 supercomputer
  - Apple II, IBM PC, Notebook PC, Palmtop, Laptop, iPhone, iPad, …

Computer Systems: hardware & software

- In the past, college computer courses tended **not** to require their students to have a comparable understanding in both fields.
- This specialisation has caused a number of serious negative results.
  - Unsuitable equipments are ordered for programmers.
  - Unsuitable software fails to exploit the performance advantages offered by revolutionary new circuits.

Computer Hardware



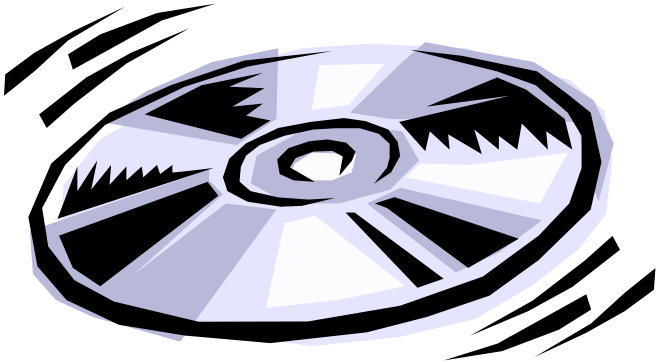
- 5 categories
  - input
  - processing
  - output
  - storage
  - communications

Computer Software

- System software
  - communication with hardware
  - resource management
  - facilitates communication among application programs

Computer Software

- Applications software
  - benefits or assists the user



Computer Systems: hardware & software

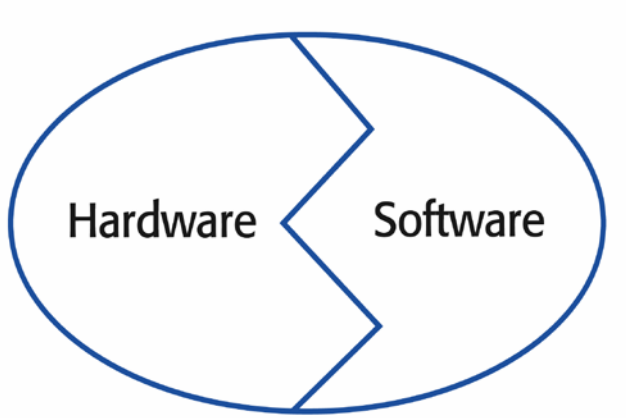


Fig. 1.2 Both hardware and software are needed.

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- We treat Computer Systems "…as a study of the interaction of hardware and software which determines the performance of … computer systems." [Wil01]

Backward (Downward) Compatibility for new hardware

- Most software written for computers with old hardware can be run on computers with newer hardware

- Why Downward Compatibility?

VHDL

- The split between those concerned with hardware and those concerned with software.
- The appearance and wide-spread using of VHDL might reverse this trend.
- VHDL.
  - Very high speed integrated circuits Hardware Description Language.
  - A programming language to be used to specify both the structure and function of hardware circuits.
  - Supports computer simulations as well as providing input to automatic layout packages which arranges the final circuits.

Computer Systems: hierarchy of systems

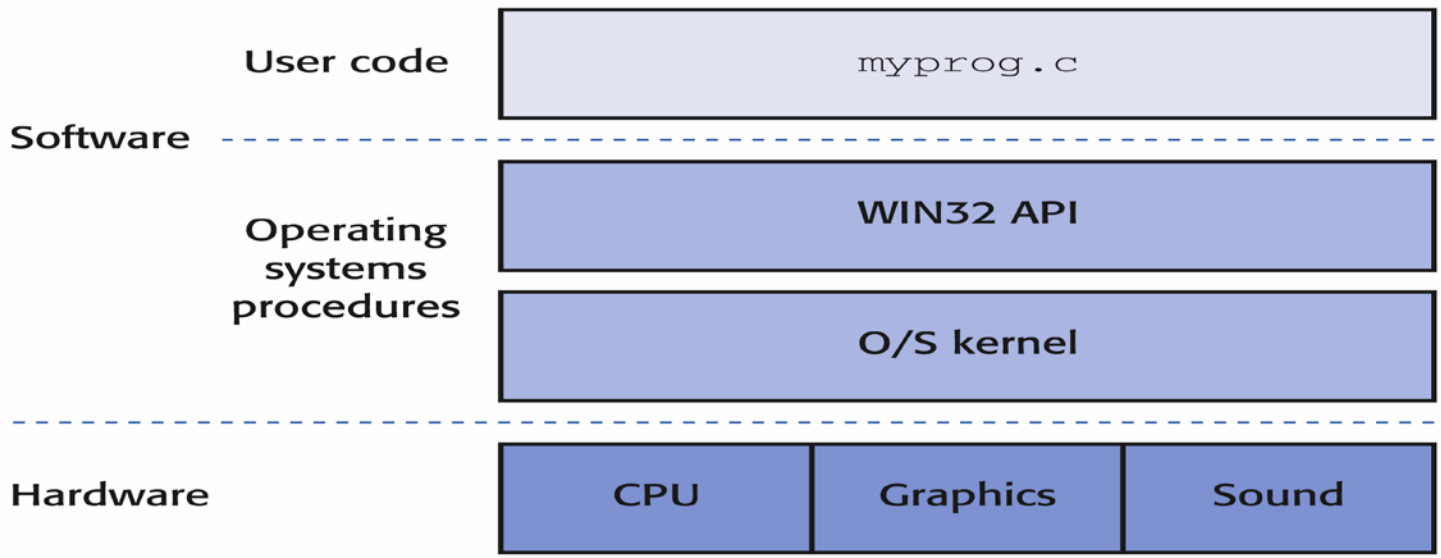

Fig. 1.3 Layers of software above the hardware.

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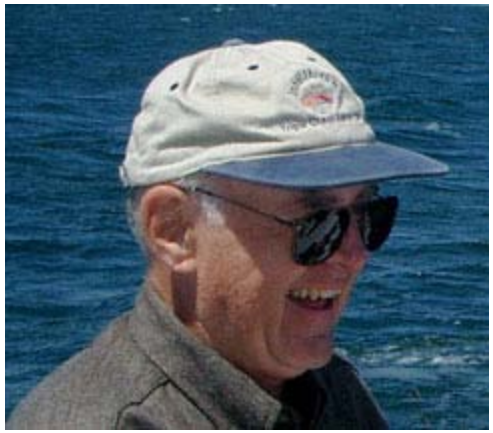
- "…computers can always be viewed as hierarchical ordered systems which can be broken down into simple component parts… in order to fully understand their operations." [Wil01]
- What are the benefits for a system to be designed in a hierarchy?

Advantages of the operating system

- Functionalities of hardware systems can be brought out by operating systems and thus offered to the user.
- The user's programs interact with hardware systems through the functionalities provided by operating systems.
  - Ease of programming.
  - Protection for the system and for other users.
  - Fairness and efficiency of using system resources

Hardware evolution: Moore's Law

- Gordon Moore (January 3, 1929 -).
  - One of the founders of Intel Corporation.
- Moore's Law (empirical).
  - The amount of circuitry (number of transistors) which can be placed on a given chip area approximately doubles every two years.



Hardware evolution: Moore's Law

Hardware evolution: Moore's Law (cont.)

- Moore's Law.
  - A circuit designed 24 months ago can now be shrunk to fit into an area of half the size.
  - It is sometimes quoted as every 18 months.
  - But Intel's official Moore's Law page states that it is every two years.
    http://www.intel.com/technology/mooreslaw/

Interacting development of hardware & software: Examples

- Windowing interfaces –WIMPs.
- A by-product of the microprocessor revolution, which allowed all users to have fast bitmapped graphics on their desks.

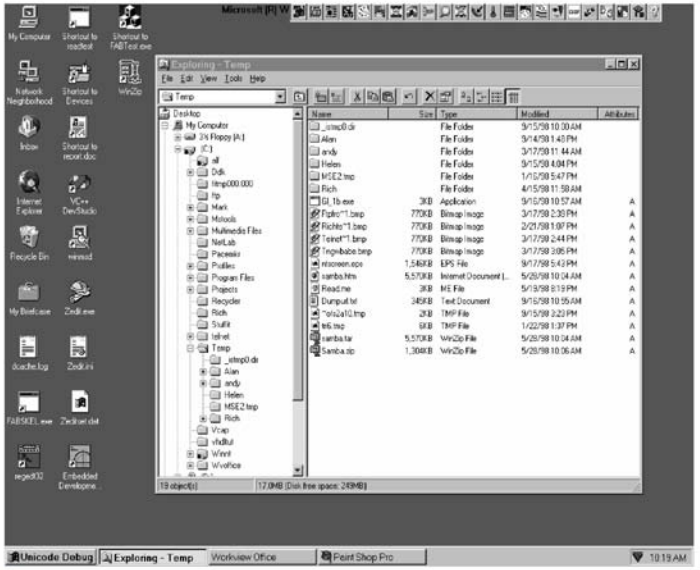


Fig. 1.7 Windows NT Explorer used as a directory browser.

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Interacting development of hardware & software: Examples

- The Internet – connecting all networks.
- The launch of Netscape browser boosted the use of Internet greatly!

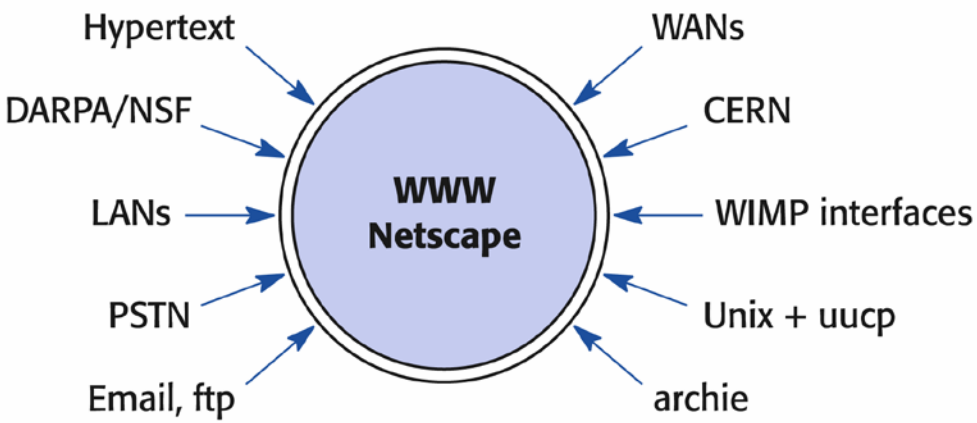

Fig. 1.9 Influences on the World Wide Web.

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Trends of Computing

- Scientific computing - computation
- Business computing - data
- Personal computing - interaction
- Pervasive computing – ubiquity
- Mobile computing - mobility

- What is next?

Q&A

- What does CPU stand for? ALU?
- What are the components of a computer system?
- Mention 4 different types of computers.
- Define ‘Downward Compatibility’.
- What are the advantages of having a hierarchical approach to computer system?
- What is Moore’s Law?
- WIMP stand for? WIMP is available due to the development of ?